CHAPTER 5 — PROCESS DISCOVERY

(How we find and model the real process)

1. What is Process Discovery

Process discovery is about **understanding how a process actually works in reality** (as-is process), not how it *should* work.

It has **four core tasks**:

1. **Define the setting**
   Decide:
   - Which process
   - Which perspective
   - Who is involved
2. **Gather information**
   Collect knowledge using discovery methods
3. **Conduct modeling**
   Draw the BPMN model
4. **Assure model quality**
   Check correctness, completeness, understandability

All four usually happen **iteratively**, not strictly one after another.

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2. Who is Involved

Two key roles:

- **Domain Experts**
  - People who actually execute the process
  - Know reality, exceptions, shortcuts
- **Process Analyst**
  - Converts knowledge into a formal process model
  - Bridges reality ↔ modeling language

👉 **Exam tip**
Domain experts know *what happens*.
Process analysts know *how to model it*.

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3. Perspective Matters

Same process looks **different from different perspectives**.

Example:

- Customer view of online bookstore
- Bookstore internal view

Both are correct, but **models will differ**.

👉 Always clarify:

- Perspective
- Boundaries
- Start and end points

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4. Challenges in Process Discovery

Challenge 1: Fragmented Knowledge

No single person knows the full process.
Each expert knows only a **piece**.

Challenge 2: Instance-Level Thinking

Experts say:

- “Every case is different”
- “We never do it the same way”

Your job:
➡ **Abstract from individual cases to general flow**

Challenge 3: Modeling Knowledge is Rare

Most experts **cannot read BPMN**.

Solution:

- Translate diagrams into **plain language**
- Validate with stories, not symbols

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5. What Makes a Good Process Analyst

- Chooses the right people
- Forms and tests hypotheses
- Identifies patterns
- Cares about model quality

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6. Process Discovery Methods

1. Document Analysis

Use:

- Policies
- Forms
- Manuals
- Organization charts

Pros:

- Cheap
- Fast

Cons:

- Often outdated
- Describes ideal process, not real one

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2. Observation

- Passive: watch people work
- Active: play a role yourself

Pros:

- Real behavior
- Captures workarounds

Cons:

- No big picture
- Observer bias

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3. Automated Discovery (Process Mining)

Uses **event logs** from IT systems.

Minimum data needed:

- Case ID
- Activity name
- Timestamp

Produces:

- Discovered model
- Conformance checking
- Enhancement

Pros:

- Objective
- Scales well

Cons:

- Needs clean data
- Only sees what is logged

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4. Interview-Based Discovery

Types:

- Structured / unstructured
- Forward / backward

Big risk:

- Exceptions are forgotten

👉 Always ask:
“What happens when something goes wrong?”

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5. Workshop-Based Discovery

- All stakeholders together
- Shared understanding
- Often fastest

Depends heavily on:

- Organizational culture
- Power distance

PROCESS MODELING METHOD (STEPWISE)

Step 1: Identify Process Boundaries

Define:

- Trigger
- Outcomes (positive + negative)
- Perspective
- Inputs and outputs

Step 2: Identify Activities and Events

- Activities = work done
- Events = something that happens instantly

Step 3: Identify Resources and Handoffs

Who performs what
Where responsibility changes

Step 4: Identify Control Flow

- Sequence
- Decisions
- Parallelism
- Loops

Step 5: Add Additional Elements

- Data objects

- Exceptions
- Message flows
- Timers

👉 Stop modeling when:

- Model answers its purpose
- More detail adds confusion, not clarity

PROCESS MODEL QUALITY ASSURANCE

1. Syntactic Quality (Verification)

Is the model **correct according to BPMN rules**?

Structural correctness

- Start events have no incoming flow
- End events have no outgoing flow
- No disconnected nodes

Behavioral correctness (Soundness)

Three rules:

1. **Option to complete**
   Process must always finish
2. **Proper completion**
   No extra tokens at the end
3. **No dead activities**
   Every activity must be executable

Typical Errors

- Deadlock → process stuck
- Livelock → infinite loop
- Dead activity → never executed

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2. Semantic Quality (Validation)

Does the model **match reality**?

Two criteria:

- **Valid** → no impossible behavior
- **Complete** → all real cases covered

Checked by:

- Domain experts
- Walkthroughs
- Scenarios

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3. Pragmatic Quality (Usability)

How useful is the model?

Depends on:

- Understandability
- Maintainability
- Learnability

Improved by:

- Clear labels
- Structured flow
- Limited size

Modeling Guidelines (7PMG)

1. Use few elements
2. Minimize routing paths
3. One start and one end per outcome
4. Structured modeling
5. Avoid OR gateways
6. Verb–object labels
7. Decompose large models
